Vansh Pratap Singh Ranawat

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Summary

Computer Science Engineering graduate focused on Backend, Cloud, and Software Engineering, with hands-on experience in Javascript, Java, REST APIs, and web development. Proficient in Git with strong foundations in OOP, Data Structures and Algorithms, and SDLC. Interested in open-source software, automation, and scalable systems.

Technical Skills

- **Languages**: Java, JavaScript (ES6+), SQL, HTML5/CSS3
  Cloud & DevOps: CI/CD Pipelines, GitHub Actions, Cloud Deployment, Vercel, Continuous Integration, Continuous Delivery, Deployment Automation
  Tools & Platforms: Git, GitHub, Postman, VS Code, Docker, REST APIs
  Backend & APIs: REST API Design, Node.js, Express.js, JWT Authentication, Backend Development, API Integration
  Database: MongoDB, SQL, Relational Databases , PostgreSQL
  Frontend: Javascript, Component-Based Architecture, Responsive Design
  Software Engineering: Object Oriented Programming (OOP), SDLC, Data Structures, Algorithms, Debugging,Problem Analysis, Code Quality, Maintainability

Projects

Receipt OCR Data Extraction System Jan 2026 – Feb 2026
Python • OpenCV • Regex • Linux • Git Workflow • CI/CD GitHub

- Engineered a modular OCR pipeline in Python to parse and extract 3 key structured data fields (store name, date, total amount) from raw receipt images into machine-readable JSON.
- Refactored image preprocessing techniques via OpenCV, applying Gaussian blurring and Otsu thresholding to improve character recognition accuracy by over 30% on noisy scans.
- Formulated regex-based validation algorithms to strictly normalize 10+ varying date formats and accurately distinguish subtotal, tax, and total amount fields.
- Integrated confidence scoring and comprehensive edge-case handling logic, driving overall extraction reliability and system robustness to 92%.
- Modularized the software architecture into 4 core reusable Python components, enabling local command-line execution and scalable batch processing of up to 100 images simultaneously.

Tamper-Proof Certificate Verification System Sep 2025 – Nov 2025
Blockchain • Solidity • Web3.js • REST APIs • Cloud Deployment Live Demo | GitHub

- Designed a decentralized engineering solution to replace manual verification pipelines with an immutable, tamper-evident cryptographic record.
- Deployed 3 Solidity smart contracts for secure on-chain storage, maintaining rigorous code quality and decentralized infrastructure standards.
- Incorporated SHA-256 cryptographic hashing to generate deterministic document fingerprints, minimizing overall verification latency from minutes to under 5 seconds.
- Configured Web3.js and MetaMask integrations to enable secure, wallet-based authentication for verifiable real-time transaction signing.
- Developed auxiliary REST APIs to issue, process, and validate credentials, rigorously stress-testing the architecture against 200+ distinct sample records.
- Enhanced smart contract execution logic through problem analysis, successfully eliminating redundant on-chain writes and lowering average gas consumption by 18%.

Lost and Found Web Application Mar 2025 – May 2025
Node.js • Express • REST APIs • MongoDB • GitHub Actions • CI/CD Live Demo | GitHub

- Architected a scalable backend-driven platform supporting 100+ concurrent users with real-time CRUD functionality and a cross-functional reporting system.
- Programmed 12+ RESTful API endpoints utilizing Node.js and Express.js, strictly adhering to MVC software design patterns and maintainability standards.
- Secured the application layer through JWT-based authentication and custom authorization middleware, successfully blocking unauthorized access across 8+ private API routes.
- Optimized MongoDB NoSQL schema design and indexed 4 frequently queried fields, drastically reducing average API response latency by 30%.
- Constructed 15+ reusable React frontend components utilizing centralized state management to eliminate code duplication and ensure long-term maintainability.
- Validated core API functionality and data integrity by engineering 25+ scripted Postman test cases, catching critical regressions prior to deployment.
- Orchestrated the CI/CD pipeline using GitHub Actions, automating build, testing, and deployment workflows and streamlining the software release lifecycle.

Education

MIT ADT University, Pune 2022 – 2026
Bachelor of Technology in Computer Science and Engineering

St. Anselm Sr. Sec. School, Bhilwara (CBSE) 2020 – 2021
Class XII — Science (PCM + Computer Science)

Certifications

Foundations of Prompt Engineering — AWS Training & Certification — July 2026 View Certificate
Introduction to Software Engineering — IBM (Coursera) — Nov 2024 View Certificate